



ME1103

Static Analysis with SOLIDWORKS

**Department of Mechanical Engineering
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1) Introduction to SOLIDWORKS

SOLIDWORKS is a brand within [Dassault Systèmes](#). It is a comprehensive software that encompasses computer-aided design (CAD) and computer-aided engineering (CAE) capabilities, which nowadays are essential in engineering product design and development. It was founded as Winchester Design Systems by Massachusetts Institute of Technology graduate Jon Hirschtick on December 30, 1993. Jon and cofounders Dr. Constantine Dokos, Scott Harris, Bob Zuffante, Mike Payne, and Tommy Li aimed to create 3D design software that was affordable, easy to use, and available on a PC running on Windows OS. Although it started off as a CAD software, for creating complex 3D models using a parametric feature-based approach, it also incorporated the CAE abilities to perform analyses and simulations, including Multi-body Dynamics simulations (SOLIDWORKS-Motion Study) and Finite Element Analysis (SOLIDWORKS-Simulation), and [many more](#).

In this exercise, we will demonstrate how to perform two types of static analyses using SOLIDWORKS, namely:

- i) Static Analysis using SOLIDWORKS-Motion study, and
- ii) Truss Analysis using SOLIDWORKS-Simulation.

The primary objective of including this exercise in the course is to create the awareness of such simulation capabilities of SOLIDWORKS, which allow one to perform engineering analyses. This complements the study of theories and concepts of basic mechanics, and it is an essential skillset for mechanical engineers in modern world. The CAD modelling part of SOLIDWORKS is covered/will be in ME2102 – *Engineering Innovation and Modelling*. Hence, the focus here is on how to create the simulation models, such as defining constraints and loadings on the models, run the simulations, and finally post-process the results. You should also be aware that the two simulation modules were developed independently, and hence their approaches to create the simulation models can be different.

1.1. Download and install SOLIDWORKS

All NUS students can download SOLIDWORKS (student version) during his/her candidature. Below are the instructions to download and install the SOLIDWORKS Student Engineering Kit (SEK) onto your computer.

- You need to log into NUS network to access the download file. If you are outside of NUS, please access this link through [webVPN](#).
- Login using your NUSNET ID: nusstu\UserID
- Click on the following link to download the Solidworks-SEK.
<https://sm05.stf.nus.edu.sg/solidworks/>
- Thereafter:
 - *Step 1*: Download installation image (ZIP) and unzip files on your computer.
 - *Step 2*: Locate the file StartSWInstall.hta and launch the file.
 - *Step 3*: Click on the "Install SolidWorks products now" button to start installation. (The SolidWorks Serial Number is embedded)

As mentioned, SOLIDWORKS is developed for Windows OS, and hence it cannot be used directly in Mac machines. Below is a possible way to overcome this limitation, but at the expense on the computer performance. <https://www.innova-systems.co.uk/solidworks-on-apple-mac-yes-we-can/>. Alternatively, you can choose to work on the simulations in NUS campus.

1.2. Working in campus

In case if you have problem installing the software or choose not to install it, you can still access SOLIDWORKS in NUS, such as the eitu PC clusters @ E2-03-06 to E2-03-09. Note that you may need your student pin number to access the premises, which can be retrieved [here](#) (if you forget about it).

2) Statics Analysis with SOLIDWORKS-Motion Study

SOLIDWORKS-Motion study is a virtual prototyping tool that supports animation, analysis and design of mechanical systems. One basic assumption here is that all the components are regarded as rigid bodies that do not deform in the analyses. Although this simulation package is designed to analyse dynamic problems, it can also be used for static analysis.

2.1. Payload of assembly in 3D

A F-shaped frame is welded onto a cylindrical collar *A*, which in turns is placed in a fixed vertical pin, as shown in Fig. 1. It can slide freely along the pin and is supported by two cables, i.e. *CF* and *DE*. Suppose cable *CF* and *DE* can only withstand 200 N and 500 N, respectively. Determine the maximum load *P* that can be applied at *C*, and the corresponding tensions in the cables and reaction forces at *A*.

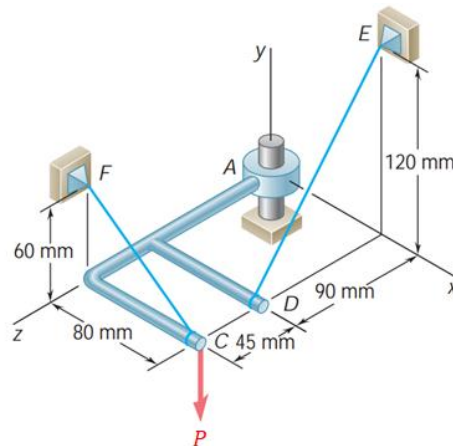


Fig. 1: F-shape frame supporting load *P*.

2.2. SOLIDWORKS-Motion Study

2.2.1. Creating the assembly model

First, download all the SOLIDWORKS files from CANVAS into a common folder, which include:

- An assembly model of the example, **Example-3-10.sldasm**.
- All the parts that are needed to build assembly model, namely: **Guiding-rod.sldprt**, **F-frame.sldprt**, **Cable-1.sldprt**, and **Cable-2.sldprt**.

Next, **double clicked** on the assembly file, i.e. **Example-3-10.sldasm**, to invoke **SOLIDWORKS**. At this point of time, only the **Guiding-rod** is present, while the rest of the components will be imported and placed in the appropriate position with respect to this fixed rod. Before that, we need to define two points in the model (corresponding to point *E* and *F* in Fig. 1) to which the cables will be attached.

To manipulate the graphic view of the model, such as rotating it, zoom in and out, etc. you can use the **view modifying toolset** as shown below.

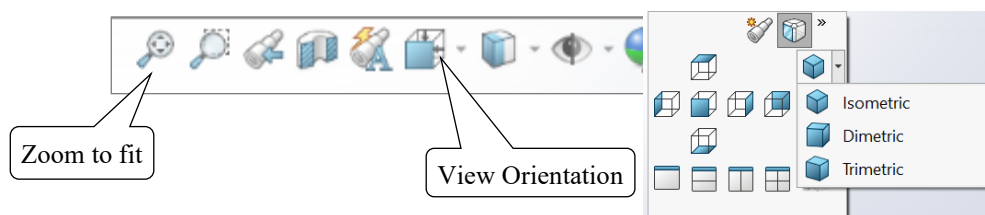


Fig. 2: Manipulating the model with **view modifying toolset**.

Note that if you are using a mouse with a roller middle button, scrolling it allows you to zoom in and out on the model, while press and holding the button rotates the view. Explore and familiarize with these operations before moving on.

The **Guiding-rod** is placed where its geometrical center coincides with the global coordinate system, indicated by the blue arrows. With this, we can define the two points on the “walls”, which falls on the **Top** and **Right planes**. To do this, choose the **Sketch** tab, and then click on the **Point** icon. This will invoke the **sketch** tool, which ask for the plane that the point will be placed.

Let us start with the point on the x - y plane. Click on the **Front Plane** and the model will orient to that view. Click on somewhere in the top-right quadrant to place the **Point**. Then use the **Smart Dimension** tool to position it at the exact location, which is 0.08 m and 0.12 m w.r.t. the global coordinate system. Finally, **Exit Sketch** to complete the step. Repeat the process to define the point on the **Right plane**, which has the coordinates of $z = 0.135\text{ m}$ and $y = 0.06\text{ m}$.

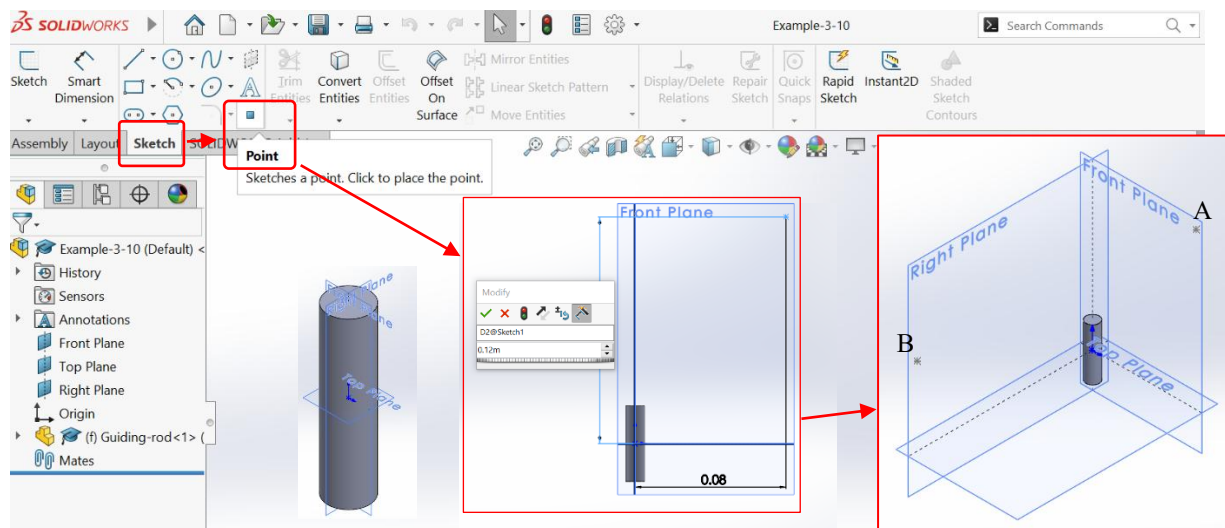


Fig. 3: Creating the fixed points on the walls.

Now, we are ready to import the rest of the components. Choose the **Assembly** tab and click on the **Insert Components**. At the **Insert Component** interface, **Browse** and choose **F-frame**, and you can drop it at anywhere in the graphic area.

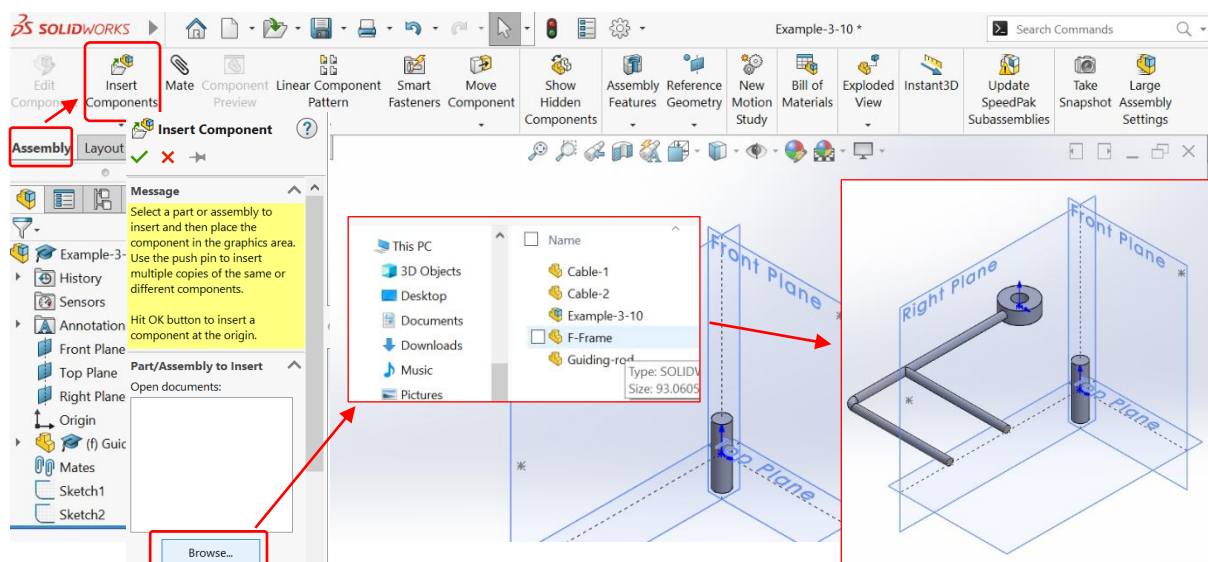


Fig. 4: Importing F-frame into the assembly model.

Next, we impose the **mate** conditions to move **F-frame** to the desired position, which simultaneously constrain it to slide along the **Guiding-rod**. At the **Assembly** tab, choose **mate**, and then at the mate interface, pick the **inner cylindrical surface of F-frame** and then the **cylindrical surface of the Guiding-rod**. With these mate selections, it will automatically recognize and assign the mate to be the **Concentric Mate Type**. Click the **Tick** icon to complete the step.

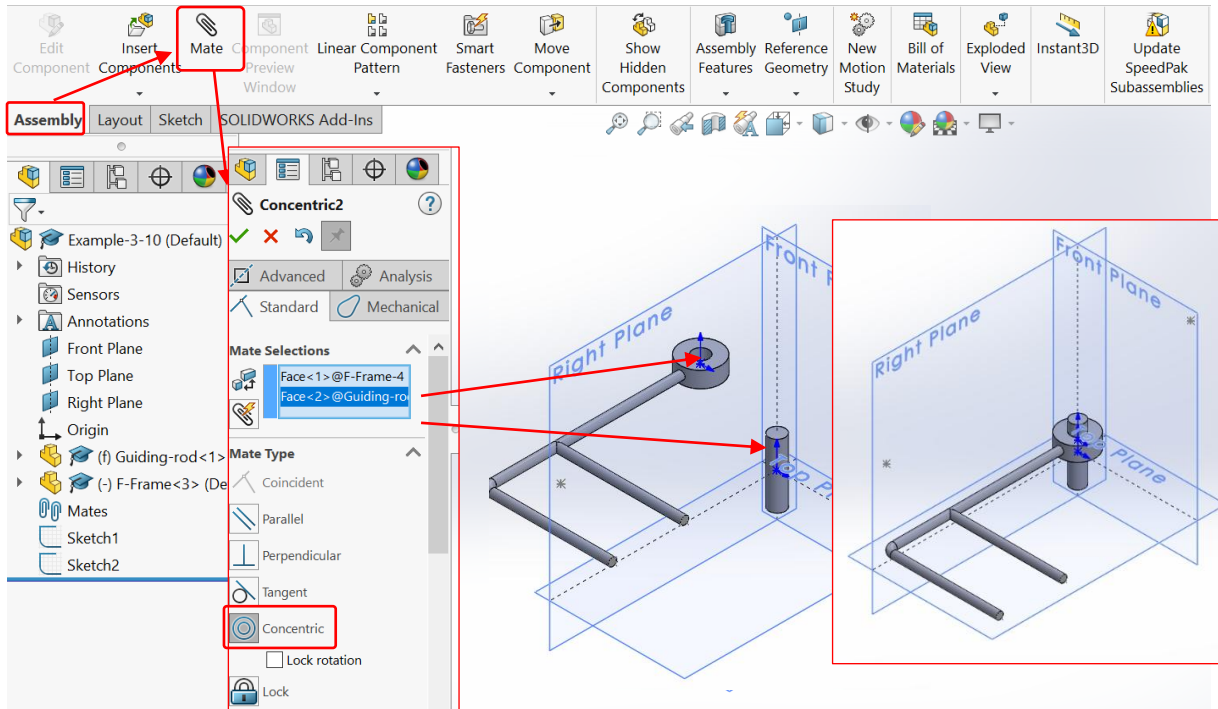


Fig. 5: Apply **mate** conditions to enforce the sliding constraints between **F-frame** and **Guiding-rod**.

In addition, we also need to position the **F-frame** such that its mid-plane coincides with the Top plane of the coordinate system. This is important so that the cables can be positioned correctly later. To do this, simply pick on the **F-Frame** and then click on **Move Component**. At the interface, choose **To XYZ Position** from the drop-down options, and set **X = Y = Z = 0**. Finally, click **Apply** to move it, and then on the **Tick** to close the **Move Component** interface.

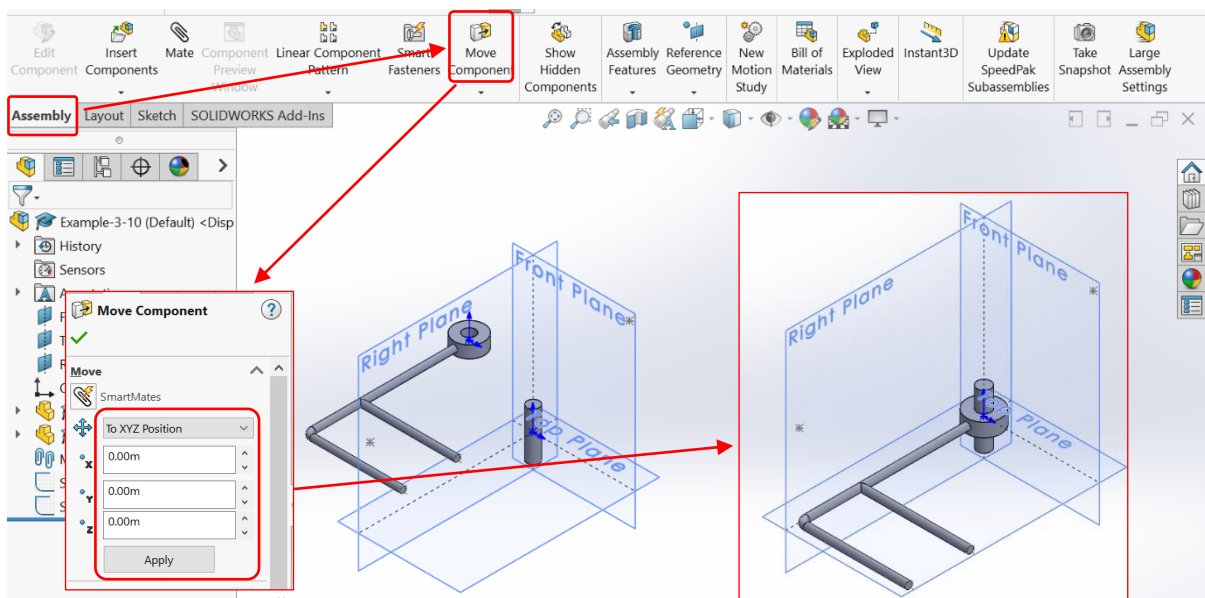


Fig. 6: Move the **F-frame** to the correct orientation before connecting it to the cables.

Finally, we can import the cables and position them in the assembly model to complete the building of the assembly model. Repeat the following step to insert the cables into the assembly model, i.e. **Assembly tab > Insert Component > Browse > Cable-1 & Cable-2 > Drop them in the graphic area**. This is then followed by the mating operations, which require the end points on the cables to be connected to the appropriate points on the **F-Frame** (at the tips of the rods) and **Point A** and **B**, as defined in the problem. In this case, the steps involve: **Assembly tab > Mate > Pick the two points mating points > Click Tick to complete the process**. Upon completion, you should see all the Mates under the Assembly Tree. With this, the assembly model is done.

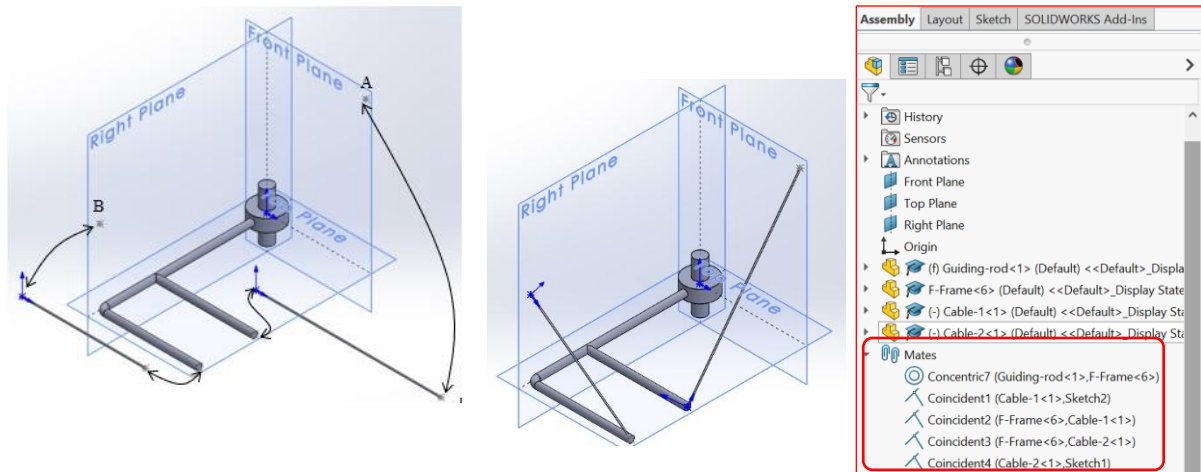


Fig. 7: Importing the cables and **mate** them appropriately to the points on the walls.

2.2.2. Static Analysis with SOLIDWORKS-Motion Study

i) Activating SOLIDWORKS-Motion Analysis solver

To perform the analysis, we need to activate the Motion Analysis module. This can be done by choosing the **SOLIDWORKS Add-Ins** tab, and then toggle on the **SOLIDWORKS Motion** icon. In case the **SOLIDWORKS Add-Ins** is not available, you can invoke this by: **Right-click on Assembly tab > Move the cursor over Tabs > Check the SOLIDWORKS Add-Ins box**.

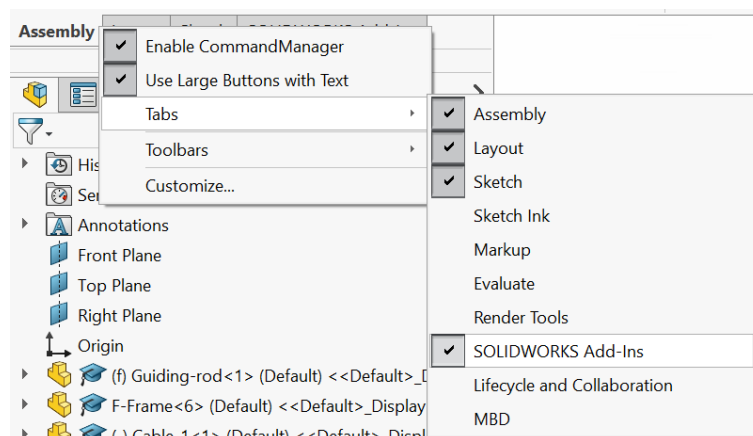


Fig. 8: Activating the SOLIDWORKS-motion study module.

ii) SOLIDWORKS-Motion Study Manager

Next, we can switch to SOLIDWORKS-Motion Study by clicking on the **Motion Study 1** (default name) tab at the left-bottom of the interface. Upon doing so, the **Motion Study Manager** pops up at the bottom half of the graphic area. This is where you defined the loadings applied on the model, then run the simulation, and finally extract the results calculated.

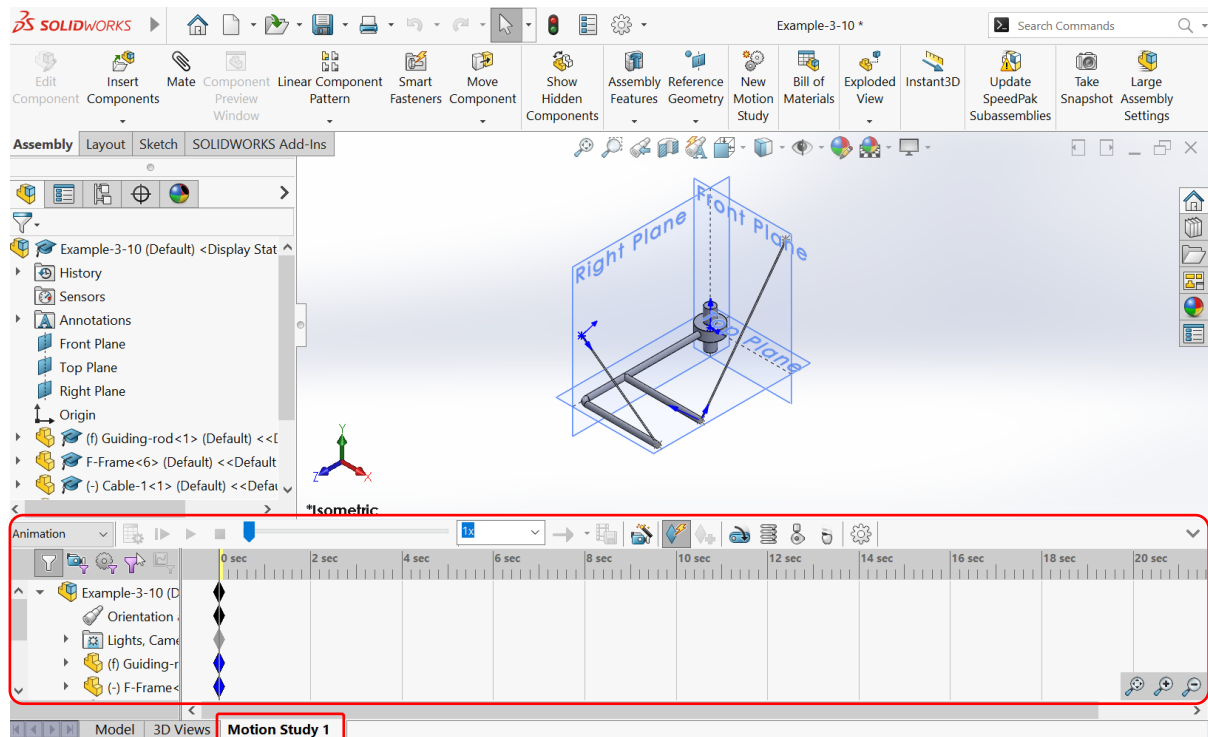


Fig. 9: Evoking the SOLIDWORKS-motion study manager.

It is mentioned here that SOLIDWORKS-Motion Study is developed to analysis dynamic problems. Hence, most of the tools provided will not be used in static analysis. Here, we will only discuss the essential ones.

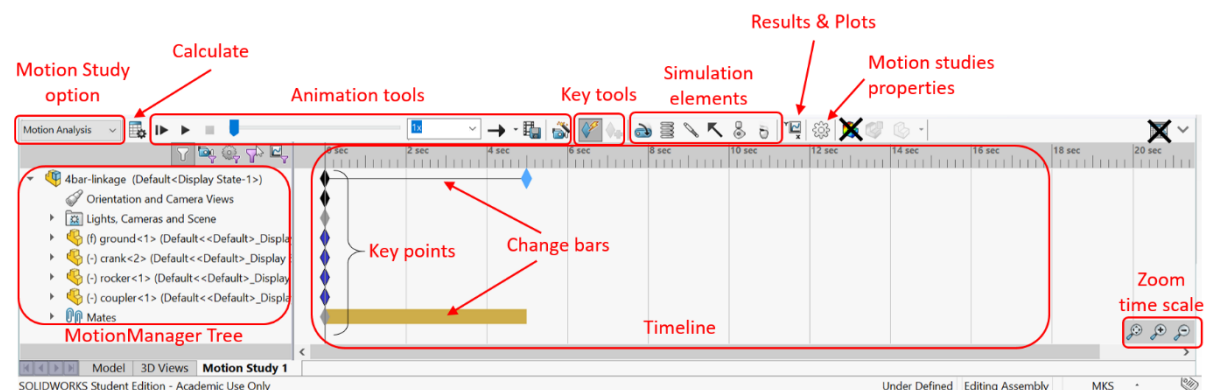


Fig. 10: Setting up the SOLIDWORKS-motion study manager for static analysis.

- **Motion Study option** – This is a drop down list, where you can choose either **Animation**, **Basic Motion** or **Motion Analysis**. Only the **Motion Analysis** option does the calculations accurately.
- **MotionManager Tree** – It contains all the information on the assembly model that you have loaded, which include the parts, sub-assemblies, mates, etc. In addition, all the other simulation information that you use in the motion study, such as spring elements, force loading conditions, and the simulation result plots, will also appear in this tree with used.

- **Calculate** – This performs the calculations of the motion simulation of the assembly.
- **Simulation elements** – These are the simulation elements that can be included into your assembly model to simulate the six specific effects:

Motor, Spring, Damper, Force, Contact, and Gravity.



Click on these icons to invoke their respective GUIs that appears at the model tree interface. You will need to provide more specific information on the type of simulation elements you have chosen.

- **Results & Plots** – This defines the types of results you want to see in your analysis, which in this study are the reaction at the cables when load P is applied. Click on the icon to bring up the GUI.

Most of the functions/icons will invoke additional GUIs for the users to specify their inputs, such as in the **Simulation elements**, and **Results & Plots**. More information on their usages will be discussed later.

iii) Applying the external loads

The next step we need to do is to applied the external loads on the model, which in this case is only load P . However, it is an unknown variable to be solved, which is constrained by the load limits imposed on the cables. To solve this problem, we can give P a representative value, say 100 N , and then check on the reaction forces induced at the cables. With the calculated reactions forces, we can then work backwards to determine the corresponding P loads at which the cables fail.

Now, click on the **Force** icon  to invoke the **Force interface**. At the **Direction** part of the interface, pick the point where load P is to be applied and **Top plane** as the direction (You can either select the **Top Plane** from the model, or from the **Assembly Tree**, which is placed on the top-left corner of the graphical area). Notice that a blue force arrow appears which is pointing downwards. In case if the direction is incorrect, **click** on the **Flip** icon next to the selection box to reverse the direction. Then at the **Force Function** part, choose **Constant** and set the force to 100 N . Note that 100 N is only a representative value for ease of calculations later. You can also choose 1 N , instead.

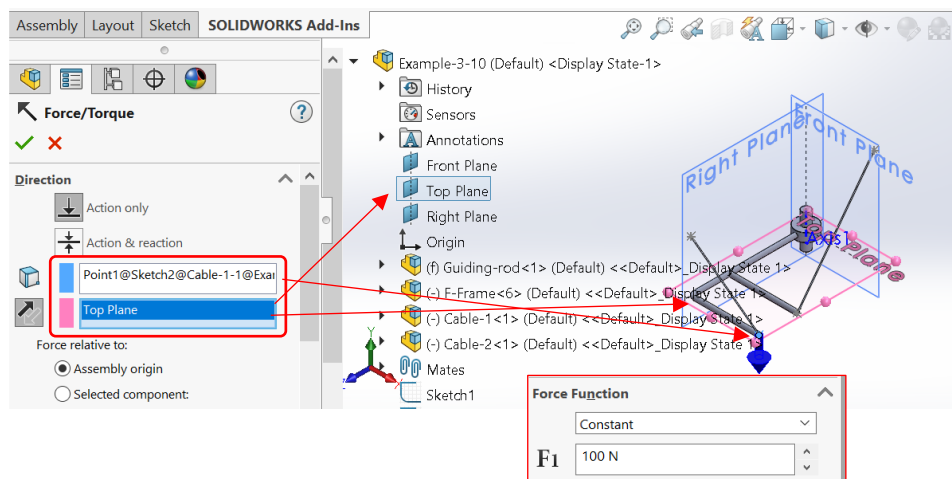


Fig. 11: Applying the external load P on **F-frame**.

With the external load P imposed, we have completed building the simulation model. The next step is simply to click on the **Calculator** icon (refer to Fig. 10) to start the solution process, which takes only a few seconds to complete.

iv) Post-processing the simulation results

Now, it is time to review the simulation results. Click on the **Results & Plots** icon to invoke the corresponding interface. At the **Result input** part, choose **Forces > Reaction Force > Magnitude** and pick **Coincident mate** under the **Assembly Tree** that represent the cable force. A force-axes appears at that point and you can use this to confirm if it is the point of concern. If this is not the correct point, **click** on the selection box and **delete** it before you can reselect another point. Finally, at the correct point, **click** on the **Tick** to complete the process.

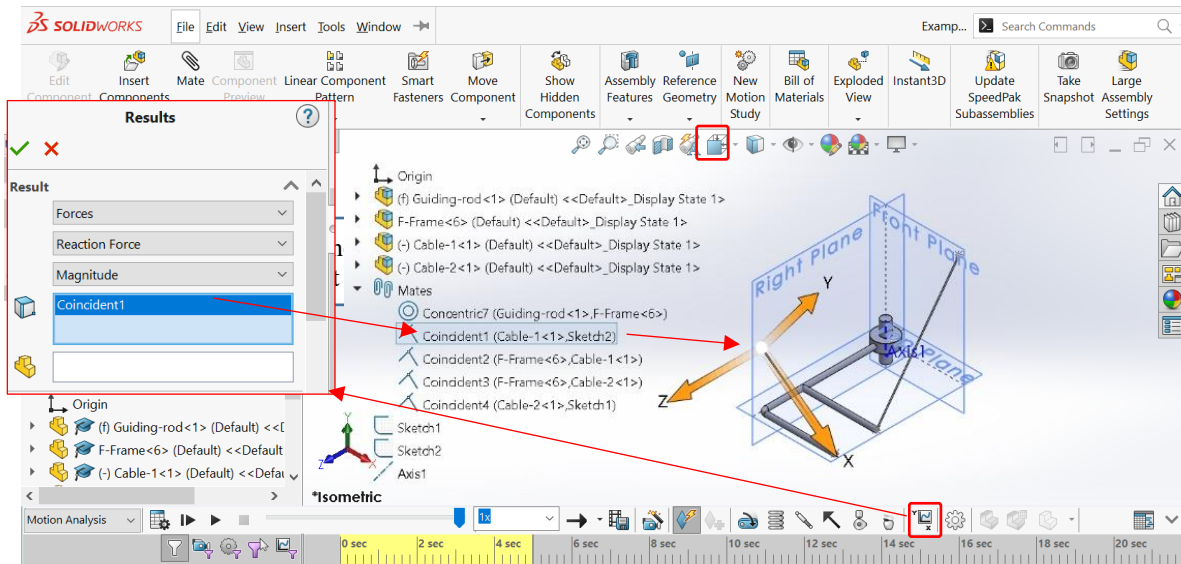


Fig. 12: Getting the tensions in the cables at post-processing step.

Upon doing so, a plot of the reaction force pops up immediately. It is no surprise that the reaction force is a constant value, since we are solving a static problem. But you may also get a linear plot if you used a different value for the P load. In either case, the results at the end of simulation time will be the ones to be used in your subsequent calculations. Notice that the range of the reaction force read between 42 to 42 N , which is inaccurate due to round-off error. **Double-click** on the **y-axis** to change the **Number** to **2 Decimal Places**. With that, the data become 41.66 ~ 41.67 N , i.e. **41.67 N** . Repeat the same process for the reaction force at the other cable, which should read **93.75 N** .

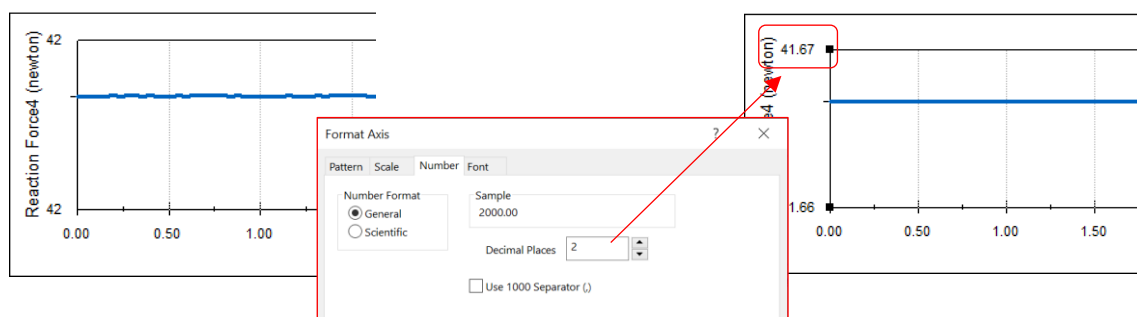


Fig. 13: Increase the decimal places of the plots for more accurate results.

Similarly, you can also extract the reaction forces at A by choosing the **Concentric** mate, which is defined as

Concentric1 (Guiding-rod<1>,F-Frame<1>)

And they are given by

$$A_x = -33.33 \text{ N}, A_y = 0 \text{ N}, A_z = -56.25 \text{ N};$$

$$M_{A,x} = 3.375 \text{ N.m}, M_{A,y} = 0 \text{ N.m}, M_{A,z} = 0 \text{ N.m};$$

Note that the forces and moments calculated above are interpreted as those acting on the first entity (Guiding-rod) due to the second entity (F-Frame where external forces act on). Hence, this implies that the supporting forces and moments the guiding rod imposes on the F-Frame are the reactions of the above values.

v) *Post-calculations*

As given in the problem, the first cable has a limiting force of 200 N, while the second one is 500 N. Suppose the first cable reach 200 N, the corresponding P load can be calculated by scaling the forces as follows:

$$P_{max,1} = \frac{200}{41.67} \times 100 \text{ N} = \mathbf{480 \text{ N}}$$

Similarly, the P load that will cause the second cable to reach its 500 N limit is

$$P_{max,2} = \frac{500}{93.75} \times 100 \text{ N} = \mathbf{533 \text{ N}}$$

Comparing the two forces, it is cleared that the limiting factor is the first cable, i.e. it will attain the 200 N limit when the P load is 480 N. And at this maximum P load, the tension in the second cable is

$$T_2 = \frac{93.75}{100} \times 480 \text{ N} = \mathbf{450 \text{ N}}$$

Finally, using the similar calculation approach above, the reaction at the concentric constrain at A are:

$$A_x = 160 \text{ N}, A_y = 0 \text{ N}, A_z = 270 \text{ N};$$

$$M_{A,x} = -16.2 \text{ N.m}, M_{A,y} = 0 \text{ N.m}, M_{A,z} = 0 \text{ N.m};$$

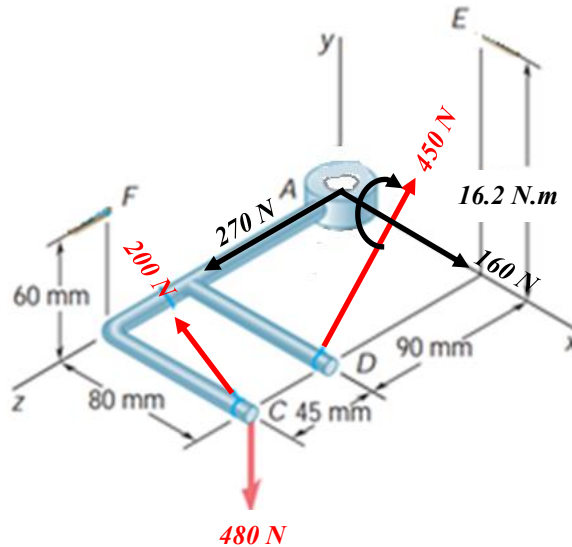


Fig. 14: Loading and supporting conditions of the problem.

3) Truss Analysis with *SOLIDWORKS-Simulation*

SOLIDWORKS-Simulation is another simulation package of *SOLIDWORKS* that uses the *Finite Element Method* (FEM) in its formulations. It is a widely used approach to solve a wide range of structural mechanical problems. Unlike *SOLIDWORKS-Motion Study*, the bodies in the analyses are deformable and hence induces stresses within the bodies when they are subjected to external loads. It is through the stress analyses that we can infer the structural integrity of the system designed.

In this exercise, we will only be looking into how to perform static analysis on truss assembly, which consists of only straight members that are joined at their ends by pin joints. And we will be using an example in your lecture notes to demonstrate it. For those who are keen to learn more about this simulation tool, *SOLIDWORKS* has built-in [self-learning tutorials](#) to assist and expediate this learning process. Enjoy.

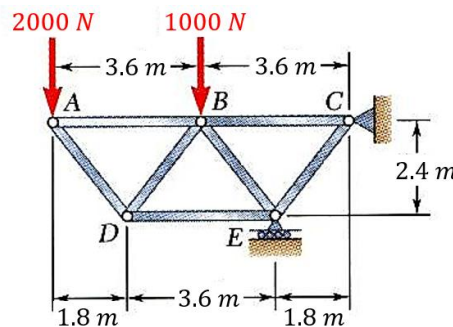


Fig. 15: Static analysis of truss structure.

Determine the forces in all the members.

3.1. *SOLIDWORKS-Simulation*

Again, download the **Truss_assembly.sldprt** from CANVAS. **Double clicked** on the file to invoke *SOLIDWORKS*. You should see a sketch of the truss assembly as multiple connected lines, which are the members in this structure. Now, we need to invoke the **Weldment tab**, in order to provide the cross-section properties for these trusses. To do that, **Right-click** on any of the tabs, and move the cursor over the **tab option**, and finally **toggle on** the **Weldments** tab. With this, the **Weldments** tab and its corresponding tools is now available.

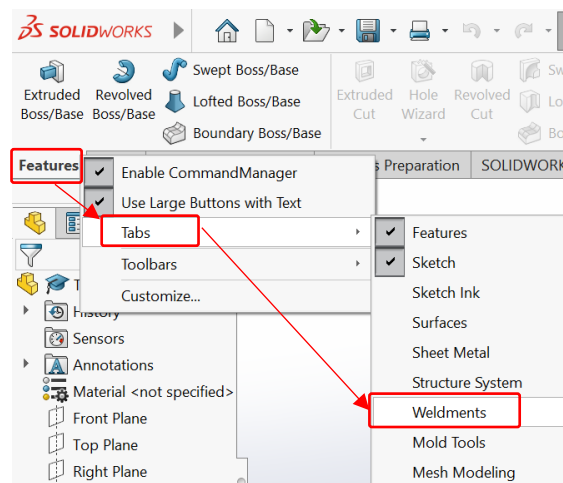


Fig. 16: Invoking the Weldment tools for truss analysis.

Now, choose the **Weldments** tab, and the **Structural Member** tool. At the input interface, choose any cross-section profile for the trusses, which in this case is of **iso standard, pipe Type** and **Size of 33.7 x 40**. We can do this because we are only interested in the member forces, and not the axial stresses induced in them. So as far as the member forces are concerned, they are invariant to the different cross-sectional profiles. On the other hand, the axial stress is simply given by $\sigma_{axial} = \frac{F}{A}$, and hence varies according to the cross-section profiles.

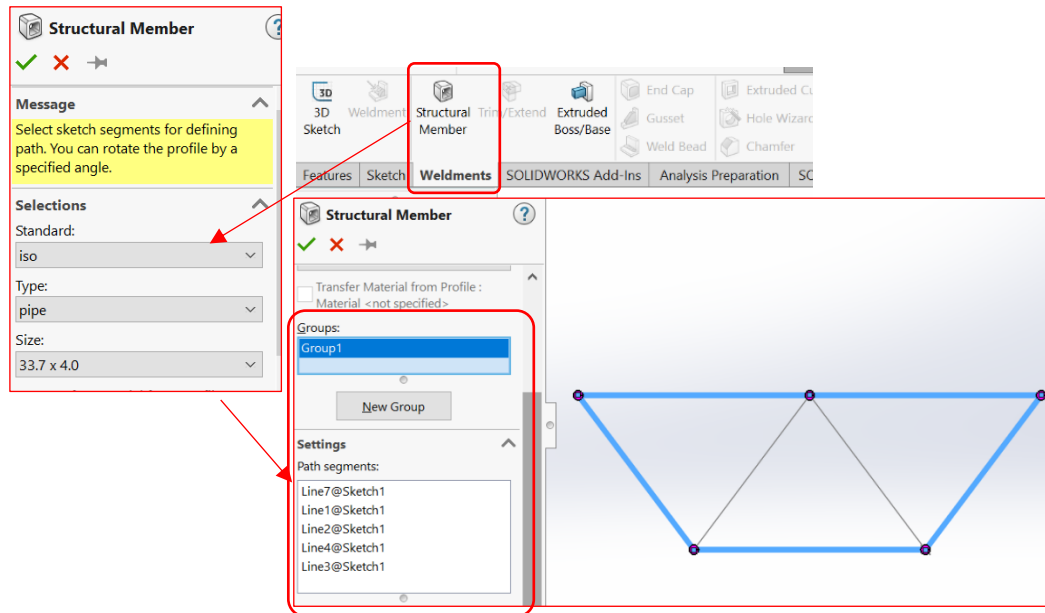


Fig. 17: Including all the members in the selection.

In the section follows, you should include all the truss members in Group1, since they share the same profile. However, SOLIDWORKS only allows connected members to be selected at one time. Hence, you may have to do it twice, such as group the five members first, then followed by the two “inner” ones. Once this is completed, you will see these pipes in the model tree.

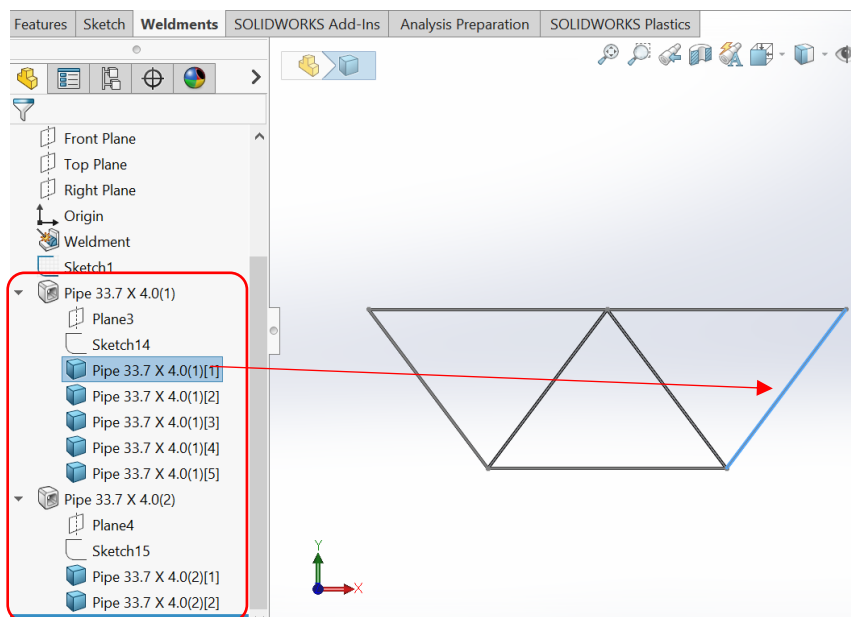


Fig. 18: Checking that all the members are included in the study.

Choose the **SOLIDWORKS-Add-Ins** tab to activate **SOLIDWORKS-Simulation**. Then click on the **Simulation** tab and **New Study** to begin the simulation. At the **Study** interface, choose **Static**, and the **Simulation tree** immediately appears beneath the model tree, and all the simulation tools beside the **New Study** are now activated. At the bottom of the graphic area, a simulation model tab (Static 1) is also created.

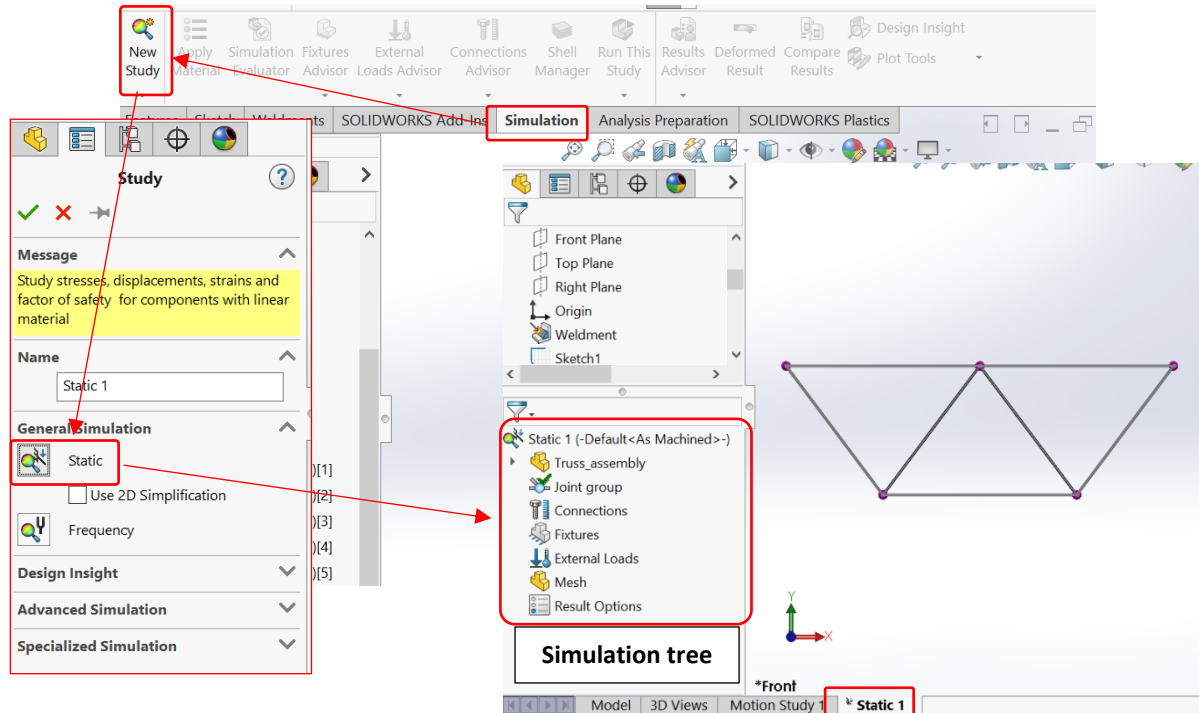


Fig. 19: Creating static study of the truss assembly.

Now, you can choose to use the simulation tools to build the simulation model, or you can do it via the simulation tree. Here, I will show the latter.

By default, **SOLIDWORKS-Simulation** treat these members as **beam**, rather than **truss**. The difference is that beam is able to take more general loads, including bending load, whereas truss can only withstand axial load. For beam, the members are assumed to be welded together at their joints, while for truss they are joined by pins. Hence, their structural behaviors are expected to be significantly different. To change them to truss, simply choose all the members (hold ctrl key and select from the list) from the tree and **right-click** to **Edit Definition**. At the **Apply/Edit Beam** interface, choose **Truss** over **Beam** option.

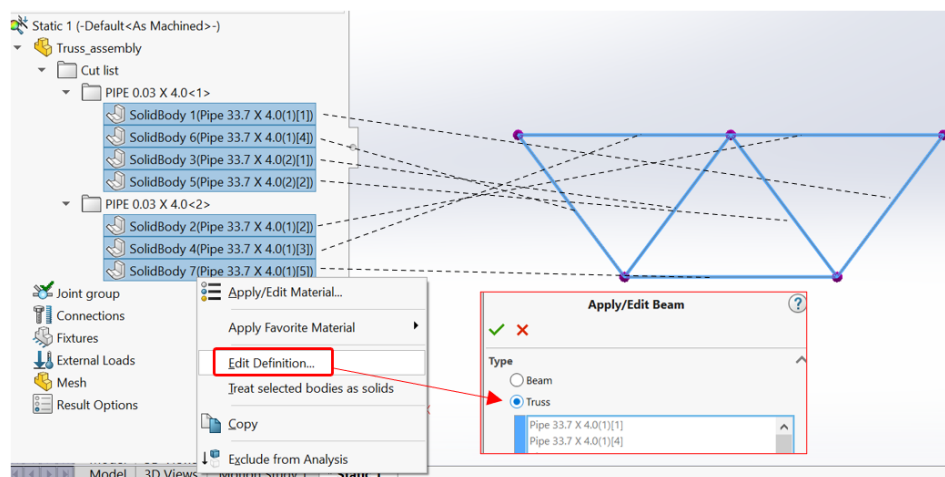


Fig. 20: Editing the members from beam to truss element type.

For the same set of members, we will define the material they are built with. Again, select all the members, **right-click** and choose **Apply/Edit Material**. *SOLIDWORKS-Simulation* provides an extensive list of materials to choose from. But since we are only interested in the member forces, the material choice is also not important. It will be important if you are concerned with the deformation of the structure. Here, we choose 1023 Carbon Steel, and remember to **Apply** for it to take effect.

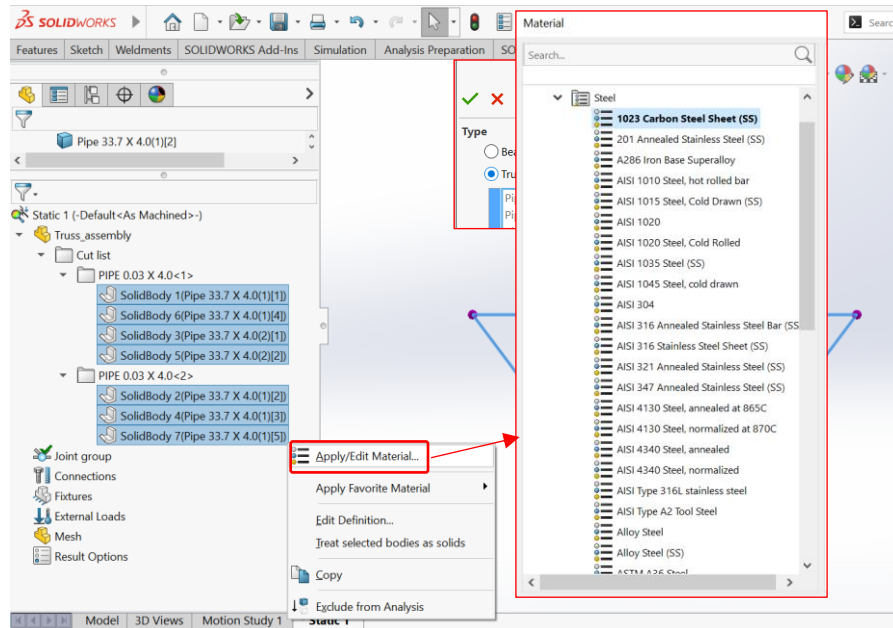


Fig. 21: Defining the material properties of the members.

Next, go to the Joint group to review the joints created for this assembly. Again, **right-click** and **Edit** to invoke the **Edit Joints** interface, and you should see five joints created for this assembly. For simple truss structures, such as those covered in this course, *SOLIDWORKS-Simulation* should be able to create the correct number of joints automatically. They are needed to impose the constraint/supporting and loading conditions on the truss, as defined in Fig. 15.

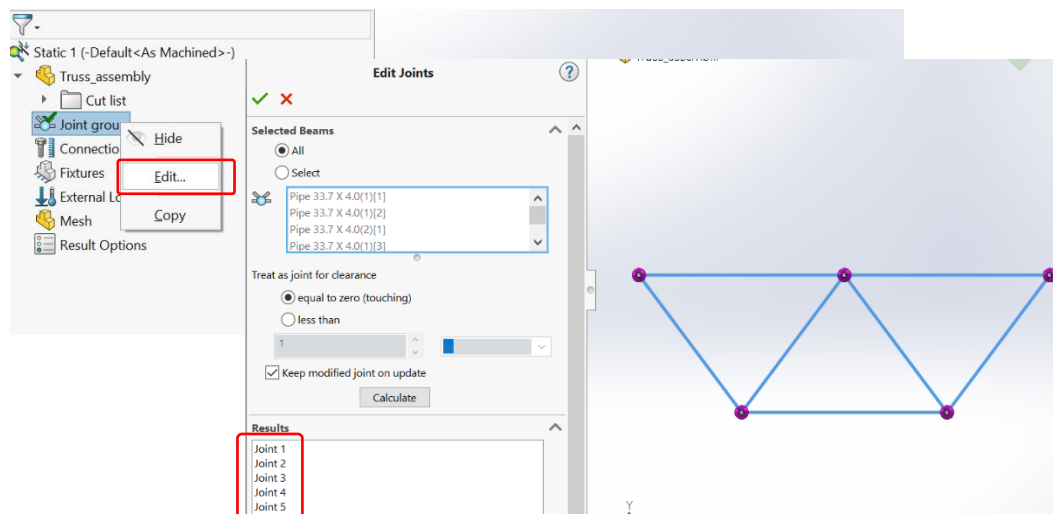


Fig. 22: Check the correctness of the joints that are automatically generated by the program.

Now, let us impose the constraints on the truss according to the supporting conditions at joints *C* and *E*. **Right-click** on **Fixtures** and then choose **Fixed Geometry**. Choose **Immovable** type for point *C* to constraint all its translation motions. You may also want to increase the **Symbol Size** to **200** so that they are clearly visible in the model.

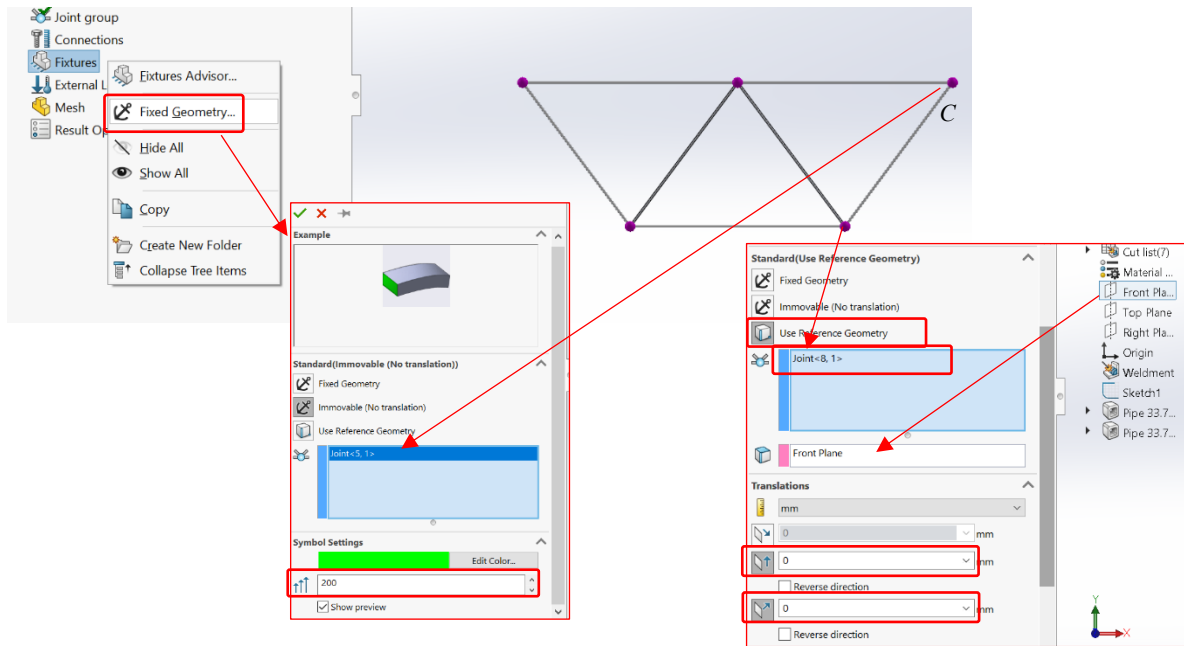


Fig. 23: Defining the required constraints on the relevant joints.

On the other hand, for point *E*, it has to be constrained using the **Use Reference Geometry** type, where the reference geometry is the **Front Plane**. And based on orientation of the **Front plane**, point *E* must be constrained in the vertical *y*-direction, and also perpendicular in the *z*-direction. Note that the constrained imposed in the *z*-direction is to limit the deformation in the *x*-*y* plane, which is essential for 2-D analysis. For the same reason, we should also impose the *z*-direction constraint on the rest of the joints.

The next step is to apply the external loads on the model, which are 2000 *N* and 1000 *N* vertically downward forces at point *A* and *B*, respectively. **Right-click** on the **External loads** and choose **Force**. At the **Force/Torque** interface, choose the **joint** type and select it from the model, and then provide the inputs as shown in the figure.

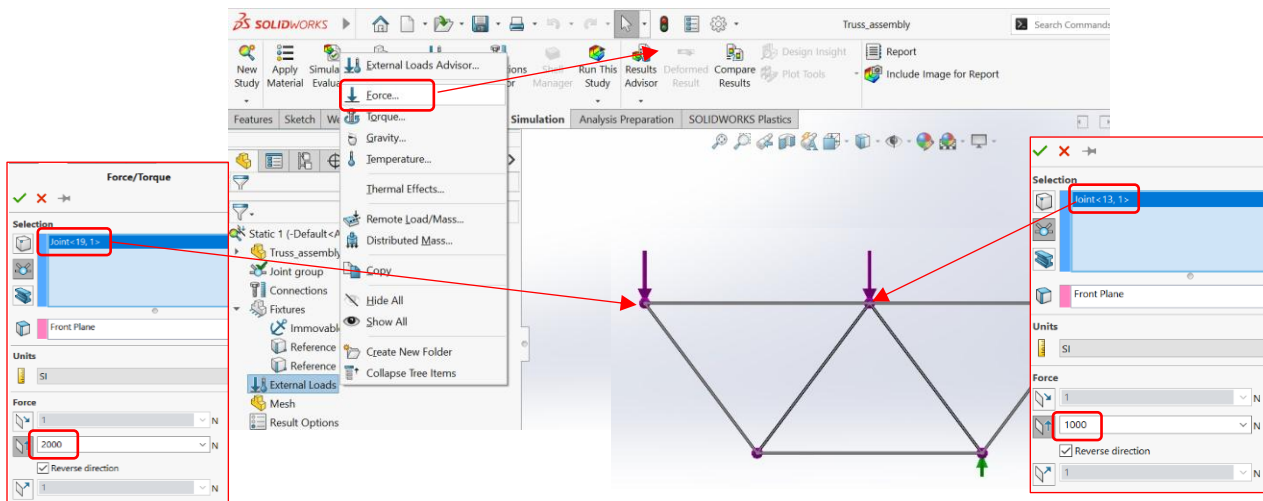


Fig. 24: Applying external loads on the relevant joints.

Finally, **right-click** on **Mesh**, and choose **Mesh and Run** to solve the problem. It takes only a few seconds to solve this simple example.

The last part of the analysis is to post-process the results. By default, the displacement and stress results will be calculated for analysis. However, in this study, neither of them matters. Instead, we are supposed to get the forces in the members. To do this, right-click on the Results and choose **List Beam Forces**.

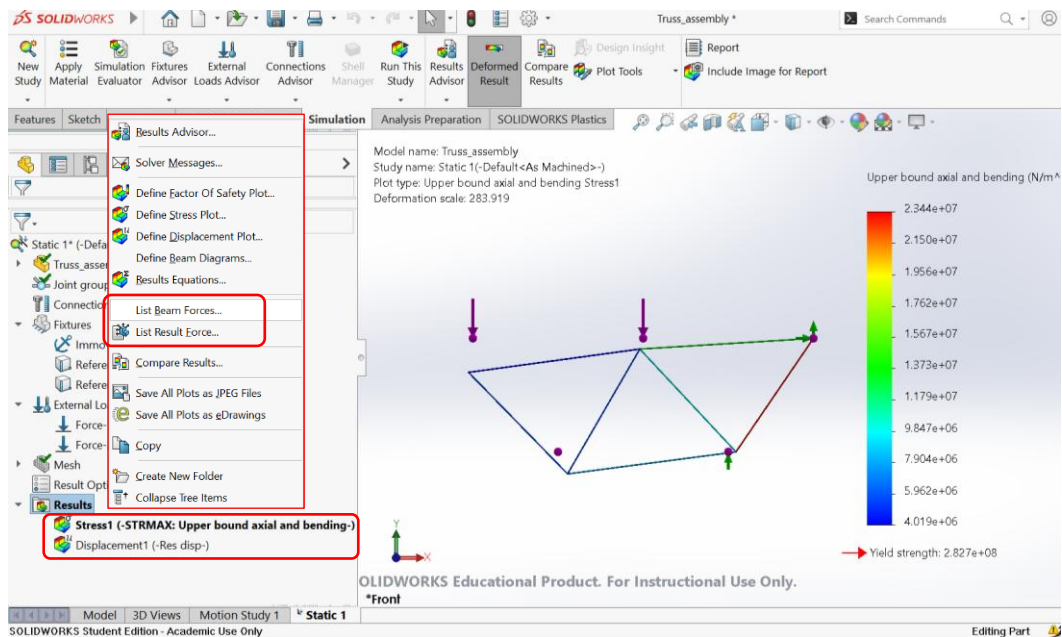


Fig. 25: Post-processing the deformed shape of truss structure.

A table listing all the member forces pops-up. Pick on the forces in the table to identify the corresponding members in the model (indicated by the beam coordinate system). And the stress state of the member is interpreted as follows. If the first force is a positive value, it signifies that the member is under compression. Otherwise, it is in tension. Hence, for member *CE*, it is in compression while for member *CB*, it is in tension. Similarly, you can also list the reaction forces at the supporting joints, by choosing **List Result Force**. A summary of the member forces is tabulated as follows, which is identical to the solution in the notes.

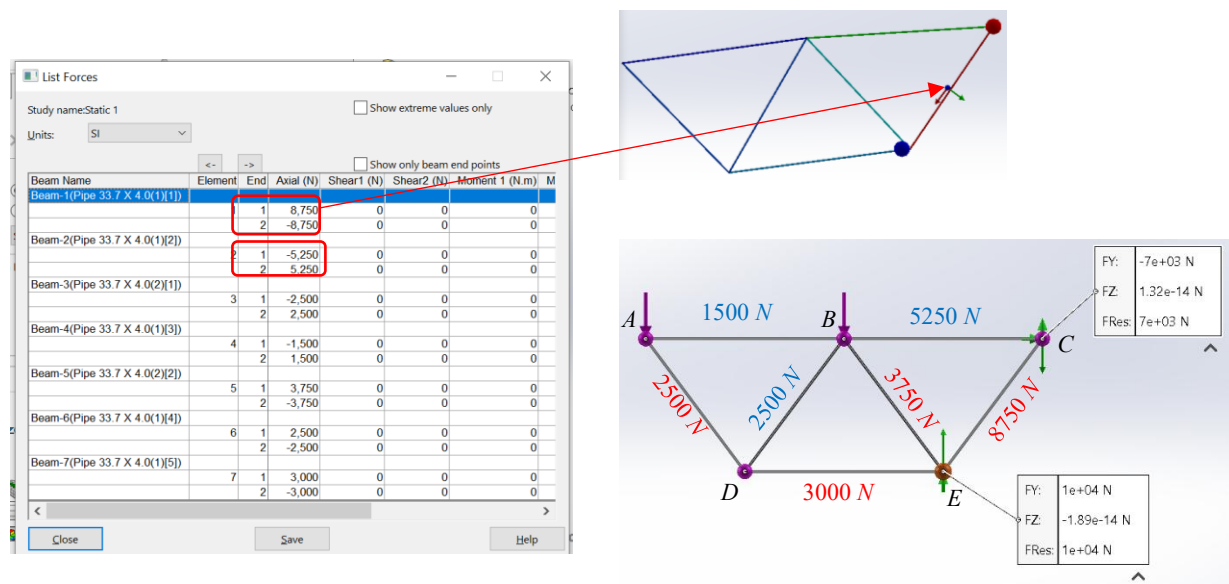


Fig. 26: Extracting the truss internal forces and reaction forces at joints.